DA storage

Recover the bytes.
Keep their meaning outside.

ELI16 · THE IDEA IN PLAIN LANGUAGE

A sealed storage locker: it preserves the package without deciding what the package means.

Why it matters: Every higher-level receipt depends on retrieving the same bytes. Keeping domain semantics outside DA makes the backend replaceable.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

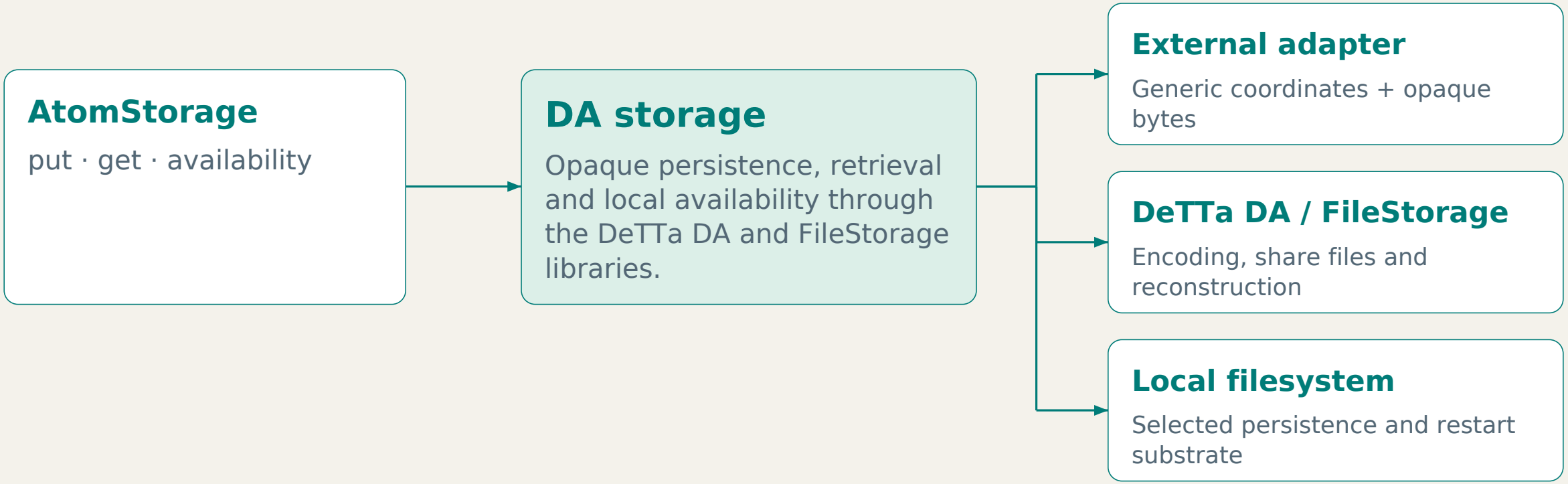
The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

Persistence	Knowledge access	Cognition	Coordination + policy
DA storage	**Relations**	**Episodes · C1**	**Messaging**
AtomStorage · C0	**Search projection**	**Beliefs · C2**	**Execution**
ConsensusFinality		**PLNKernel**	**Registry**
		Attention · C3	**Resolver**
		Inference · C4	**Authorization**

Opaque persistence, retrieval and local availability through the DeTTa DA and FileStorage libraries.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Payload and share files

Local encoded bytes and reconstruction metadata

Persistence receipts

Evidence of the selected local storage behavior

OUTSIDE THIS RESPONSIBILITY

Application interpretation

No episode identity, belief, attention, job or query semantics

Distributed custody

No validator network, indefinite retention or global availability claim

opaque-local-rs/1 · receipt detta-local-rs/1

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
put	Persist opaque bytes at generic coordinates	Returns a provider receipt
get	Recover and verify the requested bytes	Unavailable/corrupt data fails explicitly
availability	Describe observed local availability	Does not prove future retention

Local files and trusted host; recovery does not imply distributed durability.

A worked example through the boundary

REAL LOCAL PROVIDER · generated payload and injected missing files

01 · STEP

Write

Store generated bytes via the real adapter.

02 · STEP

Remove

Delete the cached payload and two erasure shares.

03 · STEP

Recover

Reconstruct identical bytes from four remaining shares.

04 · STEP

Refuse

With a third share missing, report unavailable.

What the example establishes

Availability has a failure model. Preserve it in every receipt.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Opaque boundary

No domain identifiers or numerical meaning enter the DA core.

I2

Exact bytes

Successful retrieval must match the committed payload identity.

I3

Honest availability

Insufficient reconstruction material is not a successful read.

I4

Explicit evidence

A local restart receipt never becomes a custody certificate.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Missing cache + two shares	Four of six shares recover the exact payload.	Reconstruction test
Three missing shares / corruption	Fail explicitly; do not return fabricated bytes.	Negative storage test
Process restart	Reopen local files under the same profile.	Intact-file scope only

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

4 + 2

Data + parity shares

Configured erasure profile, not measured throughput

2 lost

Shares recovered without cache

Observed reconstruction boundary

62 + 38

Upstream library tests

Clean DeTTa regression; overlapping with larger runs

No isolated read/write latency, throughput or durability benchmark is recorded. Share counts are a conformance result. Suite counts measure test execution, not performance.

PROPOSED BENCHMARK · NOT RUN

Proposed: vary payload size, missing-share count and cold/warm cache; report read/write p50/p95, bytes stored and reconstruction CPU on fixed hardware.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	Pinned DeTTa DA/FileStorage and external adapter run locally.
FIXTURE / LOCAL TEST POLICY	Generated payloads, deleted shares, corruption and injected process faults.
BLOCKED / UNSUPPORTED	No required current local gate blocked; historical dirty-source additions are outside the clean baseline.
UNRUN	Power loss, distributed custody, remote replicas, long-term retention and production load.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Measure storage tradeoffs	Controlled size/cache/failure matrix	Latency distribution and amplification per payload
Specify retention explicitly	New custody/retention contract and evidence	Expired retention cannot masquerade as availability
Explore another DA provider	Same opaque port and explicit profile	Equivalent byte/integrity conformance before adoption

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

Availability has a failure model. Preserve it in every receipt.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m11.py
```

Requires the pinned clean DeTTa source and built adapter; this is a suite command, not a microbenchmark.

`fusion/adapters/detta.py`
`adapters/detta/src/main.rs`
`docs/module-interfaces/m1_2-evidence.md`
`evidence/m5/legacy/m1_2-verification.json`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.

AtomStorage · C0

Give bytes stable identity. Own their acceptance.

ELI16 · THE IDEA IN PLAIN LANGUAGE

A library accession desk: it assigns an immutable reference and records which items were actually accepted.

Why it matters: Consumers can retry, swap providers and rebuild indexes only if reference identity and accepted-source order remain explicit.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

Persistence

- DA storage
- **AtomStorage · C0**
- ConsensusFinality

Knowledge access

- Relations
- Search projection

Cognition

- Episodes · C1
- Beliefs · C2
- PLNKernel
- Attention · C3
- Inference · C4

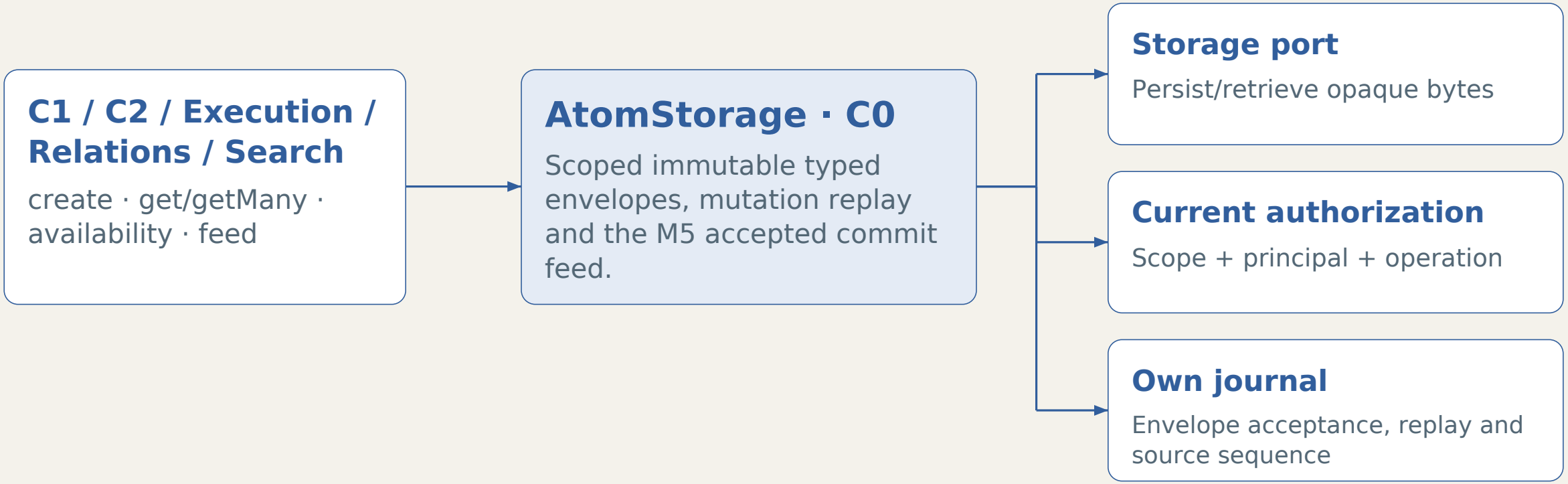
Coordination + policy

- Messaging
- Execution
- Registry
- Resolver
- Authorization

Scoped immutable typed envelopes, mutation replay and the M5 accepted commit feed.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Envelope + replay identity

Immutable reference, content type and request semantics

Accepted source feed

Source UUID, scope and contiguous sequence in the M5 profile

OUTSIDE THIS RESPONSIBILITY

Knowledge meaning

No query interpretation, belief admission or relation acceptance

Cross-owner transaction

DA bytes may commit before envelope acceptance; clients recover through ports

Historical immutable-bytes/1; fresh immutable-commit-feed/1 · contract 1.1.0

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
create	Accept one scoped immutable envelope	Same intent replays; changed intent conflicts
get / getMany	Retrieve exact typed bytes for references	Live authority and integrity checks
availability	Report selected provider evidence	Preserve provider/durability distinctions
head / readCommits	Enumerate accepted source prefix	Fresh immutable-commit-feed/1 profile

M5 caps 512 accepted atoms and 4096 bytes/atom; legacy limits remain profile-specific.

A worked example through the boundary

M5 REAL DeTTa + OWNER JOURNAL · one deliberate feed-acceptance kill

01 · STEP

Persist bytes

Write through the opaque storage port.

02 · STEP

Accept

Commit envelope and feed metadata together.

03 · STEP

Lose reply

Kill after acceptance, before caller acknowledgment.

04 · STEP

Replay

Recover one reference and one source sequence.

What the example establishes

Accepted identity is authoritative; indexing is derived.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Immutable identity

An accepted reference binds scope, revision and exact content.

I2

Replay semantics

Same principal/scope/key with changed meaning conflicts.

I3

Contiguous accepted feed

Acceptance and sequence metadata commit together; orphans do not enter the feed.

I4

Provider neutrality

The consumer does not inspect concrete provider internals.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Bytes persist; acceptance not yet committed	Retry recovers bytes, then completes owner acceptance.	Independent commit boundary
Acceptance commits; reply lost	Replay returns the same logical acceptance.	M5 feed kill case
Provider substitution	Select an independent same-profile implementation on a fresh store.	No populated-state migration

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

2 × 2

Providers × bindings

M1.1/M1.2 workflow substitution matrix

1

M5 feed SIGKILL case

Acceptance before response

512 / 4096 B

M5 atoms / per-atom limit

Contract bounds, not demonstrated scale capacity

The substitution matrix establishes matching logical outcomes. Existing whole-workflow timings include storage, messaging, workers and retries; they are not AtomStorage latency.

PROPOSED BENCHMARK · NOT RUN

Proposed: create/get/getMany across provider, payload and batch size; separate cold open, warm read and replay, with p50/p95 plus storage amplification.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	DeTTa-backed AtomOwner and fresh FeedAtomOwner with their own journals.
FIXTURE / LOCAL TEST POLICY	Independent atomic-file AtomStorage and feed; generated bytes and hostile ports.
BLOCKED / UNSUPPORTED	No required M5 gate blocked. Replacement of an already populated namespace is unsupported.
UNRUN	Power loss, distributed consistency, pruning/compaction and live namespace migration.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Quantify provider overhead	Fixed corpus and equivalent durability labels	Separate lookup latency from DA reconstruction
Design retention/pruning	Source-prefix and cursor invalidation contract first	No silent holes in accepted history
Add a production replacement	Independent owner transitions + conformance	Same consumer, explicit profile, no hidden migration

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

Accepted identity is authoritative; indexing is derived.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m5.py --gates storage --environment <env>
--output <fresh>
```

Build pinned native providers first. Storage gate uses real DeTTa; fixture passes cannot substitute for it.

`fusion/atoms.py`
`fusion/atom_feed.py`
`fusion/adapters/fixture_feed.py`
`docs/module-interfaces/m5-contracts.md`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.

Relations

Make structure explicit. Accept it independently.

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Labelled links between records: who depends on what, which role each endpoint plays, and which links were accepted.

Why it matters: A byte store cannot answer structural questions. Explicit ordered roles allow traversal without hiding semantics in payload conventions.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

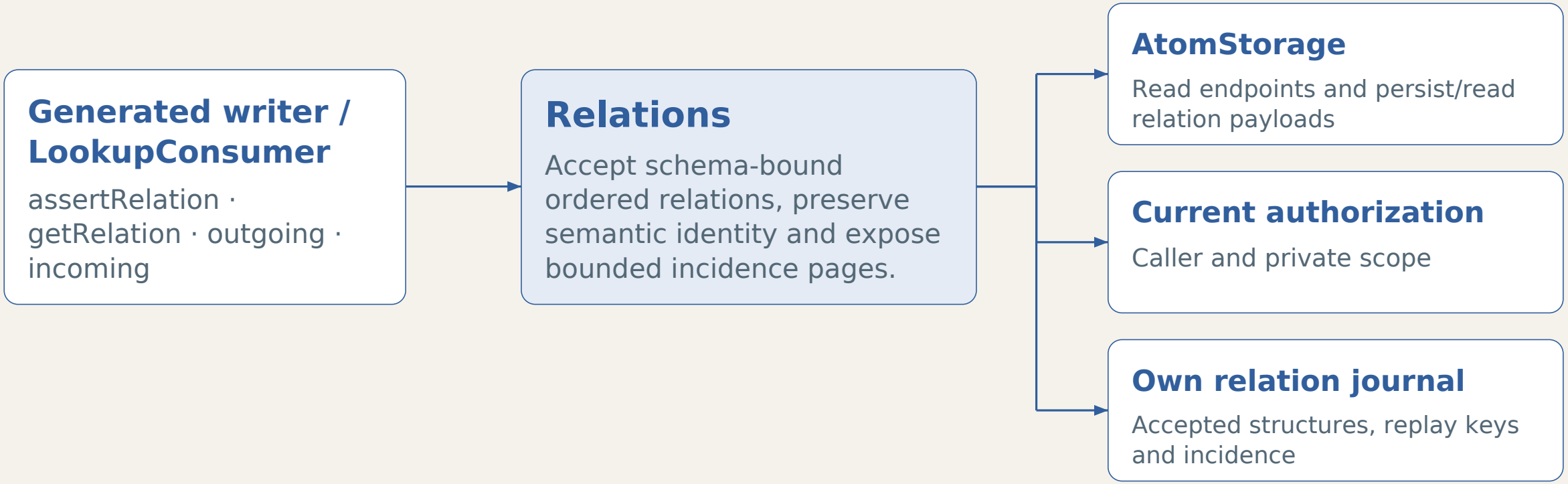
The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

Persistence	Knowledge access	Cognition	Coordination + policy
DA storage	**Relations**	Episodes · C1	Messaging
AtomStorage · C0	Search projection	Beliefs · C2	Execution
ConsensusFinality		PLNKernel	Registry
		Attention · C3	Resolver
		Inference · C4	Authorization

Accept schema-bound ordered relations, preserve semantic identity and expose bounded incidence pages.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Accepted structural ledger

Schema, direction, ordered roles and acceptance sequence

Replay + incidence

Semantic duplicate handling and fixed-head traversal

OUTSIDE THIS RESPONSIBILITY

Unaccepted payloads

An orphan atom is not an accepted or traversable relation

Truth and general graph search

No inference, ontology engine, transitive query or hidden graph backend

schema-ordered-relations/2 · contract 2.0.0 · fresh stores

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
assertRelation	Validate structure, persist payload, accept	Semantic identity excludes transport key
getRelation	Read an accepted structure	Reject orphan or malformed relation payload
outgoing	Page origin-role incidence	Fixed head and bound cursor
incoming	Page destination-role incidence	Reverse incidence; symmetric schemas explicit

128 accepted relations; 512 replay keys; pages ≤8; finite schema set.

A worked example through the boundary

ACTUAL NATIVE HIVE LOOKUP · generated private corpus

01 · STEP

Discover root

Search returns a task reference.

02 · STEP

Follow outgoing

Read its requires relations at a fixed head.

03 · STEP

Inspect roles

Resolve dependent → prerequisite identities.

04 · STEP

Verify incoming

Confirm reverse incidence; retrieve payloads.

What the example establishes

A relation becomes visible when its owner accepts it.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Structural identity

Type/schema/direction/ordered roles/scope define meaning.

I2

One incidence

Self-links and repeated endpoints do not duplicate the same incidence.

I3

Snapshot pagination

Cursor binds principal, scope, endpoint, direction and accepted head.

I4

Independent acceptance

Payload creation and relation acceptance remain separate commits.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Crash after payload, before acceptance	Orphan stays invisible; retry may accept it.	One real process-kill case
Crash after acceptance, before reply	Recover the accepted relation once.	One real process-kill case
Changed cursor / revoked authority	Fail explicitly; no cross-scope continuation.	Malformed/cursor/auth tests

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

4 / 4

Logical operations exercised

Contract + real-provider + native consumer

15

Requires edges in native demo

Five roots connected to three prerequisites

128

Accepted relations per fresh store

Declared cap; no throughput claim

The 15-edge example demonstrates one-hop traversal and reverse incidence. No isolated relation throughput, latency or large-graph scaling benchmark has been run.

PROPOSED BENCHMARK · NOT RUN

Proposed: vary fan-out, schema arity, repeated endpoints and page size; measure append and traversal p50/p95 with fixed-head correctness under concurrent appends.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	SQLite RelationOwner over real local DeTTa; actual authenticated Hive traversal.
FIXTURE / LOCAL TEST POLICY	Generated schemas/corpus, malformed responses and injected faults; independent storage alternative.
BLOCKED / UNSUPPORTED	No required M5 gate blocked; historical tuple Relations declaration is not silently upgraded.
UNRUN	Large graphs, distributed writers, multi-hop query planning, migration and graph-database substitution.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Extend schema vocabulary	Versioned role semantics and negative tests	No ambiguous direction or structural identity
Measure fan-out costs	Controlled degree/arity/page matrix	Stable pagination with bounded memory
Explore a second provider	Independent accepted-state implementation	Same contract outcomes through unchanged consumer

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

A relation becomes visible when its owner accepts it.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m5.py --environment <env> --chain-build <chain> --output <fresh>
```

The all-gates run separates fixtures, real storage, native Hive and legacy regressions. Keep generated private corpus and journals private.

`fusion/relations.py`
`fusion/lookup_contracts.py`
`fusion/lookup.py`
`tests/test_m5.py`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.

Search projection

Discover references.
Expose how fresh the answer is.

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A library catalogue with a visible indexing date: “nothing matches” differs from “the catalogue is behind.”

Why it matters: Consumers must discover answers without receiving answer references. A replayable source and explicit watermark make that discovery inspectable and rebuildable.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

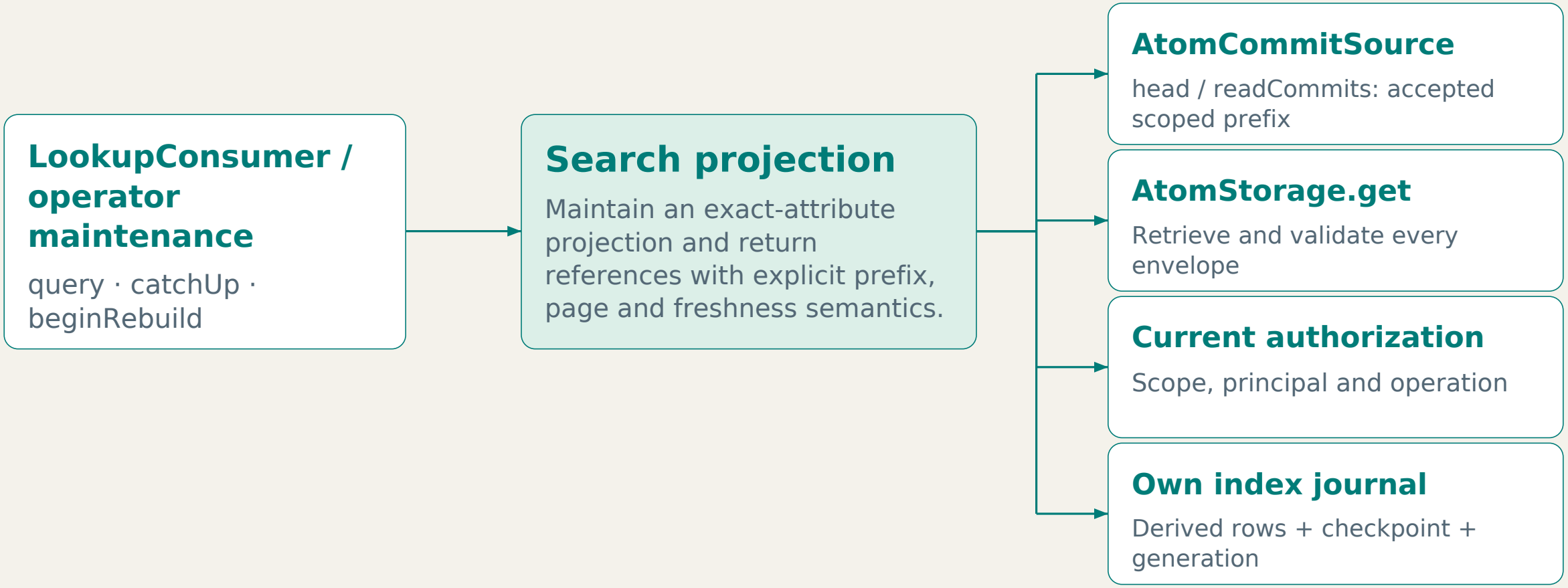
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DA storage	Relations	Episodes · C1	Messaging
AtomStorage · C0	**Search projection**	Beliefs · C2	Execution
ConsensusFinality		PLNKernel	Registry
		Attention · C3	Resolver
		Inference · C4	Authorization

Maintain an exact-attribute projection and return references with explicit prefix, page and freshness semantics.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Derived projection

Extracted exact attributes; disposable and rebuildable

Progress and generation

Atomic index/checkpoint updates, rebuild intent and cursor generation

OUTSIDE THIS RESPONSIBILITY

Authoritative source

Acceptance and immutable bytes belong to AtomStorage

Semantic cognition

No embeddings, scoring, inferred tasks or general knowledge search

exact-attributes/1 · knowledge-attributes/1 · contract 2.0.0

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
query	Exact conjunction of 1–3 distinct attributes	kind/topic/label; equality only
catchUp	Process ≤8 new source entries	Checkpoint advances only after complete batch
beginRebuild	Capture source head; create generation	Queries unavailable until captured prefix complete

256 indexed records; page/batch ≤8; query ≤1024 bytes; no automatic repair.

A worked example through the boundary

REAL HIVE + SQLITE / DeTTa · independent file Search remains a fixture

01 · STEP

Initial discovery

Five matches, three pages at limit two.

02 · STEP

Append a match

Requested barrier exceeds the indexed prefix.

03 · STEP

Signal lag

query returns NOT_AVAILABLE; no ingestion.

04 · STEP

Catch up / rebuild

Six roots and nine payloads; replacement agrees.

What the example establishes

An answer is useful only when its freshness means something precise.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Exact matching

Case/Unicode preserved; missing attributes never match.

I2

Complete prefix

No advance across gaps, unavailable bytes or forged metadata.

I3

Bound continuation

Cursor fixes query, principal, source, snapshot and generation.

I4

Honest freshness

Prefix completeness, page completion, requested minimum and observed head are distinct.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Before/after checkpoint commit	Rows and checkpoint recover together.	Both providers; four direct kills
During rebuild intent	Resume bounded rebuild; reject old-generation cursor.	Both providers; two direct kills
Native page/result crash	Resume caller progress without seeded answers.	Both providers; four native kills

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

5 → 6

Matching roots before/after catch-up

Native discovery counts, not a relevance score

2

Independent Search providers

SQLite owner + atomic-file fixture

256 / 8

Indexed records / max page or batch

Finite contract limits, not scaling results

Native tests compare complete reference sets and canonical payload bytes across four transitions. No query-latency, ingestion-rate or relevance benchmark is recorded.

PROPOSED BENCHMARK · NOT RUN

Proposed: vary corpus size, selectivity, page size and index lag; report p50/p95 query latency, catch-up rate and rebuild time with explicit freshness barriers.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	SQLite SearchOwner over accepted real DeTTa bytes; actual native Hive query path.
FIXTURE / LOCAL TEST POLICY	Independent atomic-file Search with its own predicate evaluator; generated corpus/oracle.
BLOCKED / UNSUPPORTED	No required M5 gate blocked; general search and migration are outside this profile.
UNRUN	Embedding relevance, large-corpus scale, distributed indexing, compaction and live migration.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Measure freshness/latency tradeoff	Frozen corpus, source barriers and controlled lag	Latency and staleness reported together
Broaden query language deliberately	New explicit profile and sound cursor semantics	No silent change to exact equality
Scale rebuild machinery	Capacity/retention contract before optimization	No skipped source ranges or query-time ingestion

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

An answer is useful only when its freshness means something precise.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m5.py --gates hive --environment <env> --output <fresh>
```

Requires pinned native providers. The writer exits and the oracle is removed before native lookup. Fresh replacement rebuilds derived state; it is not migration.

`fusion/search.py`
`fusion/adapters/fixture_search.py`
`fusion/lookup_contracts.py`
`tests/test_m5_boundaries.py`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.

EpisodicMemory · C1

Keep what happened.
Keep disagreement in the record.

ELI16 · THE IDEA IN PLAIN LANGUAGE

A lab notebook: a repeated report does not become a second event, and an inconvenient observation is not erased.

Why it matters: Later belief changes need a durable history of occurrences and provenance. Processing receipts must work even when no numerical belief can be formed.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

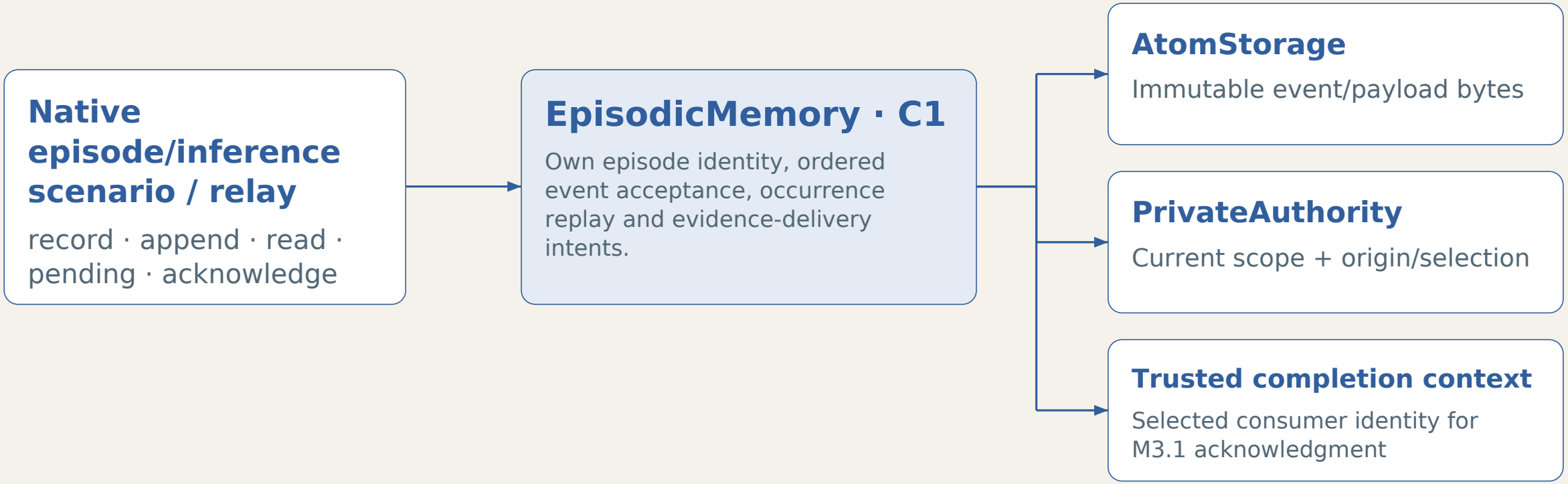
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ConsensusFinality		PLNKernel	Registry
		Attention · C3	Resolver
		Inference · C4	Authorization

Own episode identity, ordered event acceptance, occurrence replay and evidence-delivery intents.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Episode ledger

Identity/head, accepted events, occurrence semantics and replay

Delivery intent

Pending evidence and trusted model-independent completion

OUTSIDE THIS RESPONSIBILITY

Belief and numeric ownership

C2 owns admission; RevisionMath/PLN supply proposals

Foreign owner state

No C2 SQL reads and no interpretation of PLN snapshot internals

episode-local-sequence/2; M3.1 local 3.0.0 adds episode-delivery-completion/1

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
recordEpisode / appendEvent	Create identity; append at expected sequence	Event + delivery intent commit atomically
readEpisode / readEvidence	Read ordered accepted history	Observed time is metadata, not ordering authority
pendingEvidence	Return bounded pending deliveries	Backpressure prevents dropped evidence
acknowledgeDelivery	Accept trusted processing completion	M3.1 local v3; old v2 remains explicit

256 events/episode; 16 pending deliveries; bounded pages; trusted local placement.

A worked example through the boundary

M3.1 REAL DeTTa / native PLN · generated observations; historical source labelled

01 · STEP

Append 16

Accept a full batch of observations.

02 · STEP

Drain 16

Relay validates C2 receipts; C1 clears pending.

03 · STEP

Repeat

Append and drain another sixteen.

04 · STEP

Finish 1

33 retained events; zero pending after each drain.

What the example establishes

Processing completion is not the same as numerical agreement.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

One occurrence

Same occurrence/body replays; altered meaning conflicts.

I2

Ordered acceptance

Expected-head CAS controls append order; observed time cannot retime it.

I3

Atomic intent

An accepted evidence event has a delivery intent in the same C1 commit.

I4

Independent completion

Processing may finish with no numerical truth value; history remains retained.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Payload saved; append not committed	Retry through AtomStorage, then commit episode state.	No cross-owner transaction
C2 decision committed; acknowledgment lost	Relay recovers exact decision and retries trusted completion.	No manufactured acknowledgment
Capacity or malformed receipt	Pause with pending evidence; never drop or falsely complete.	Repeated-drain and hostile-receipt tests

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

33

Observations in one episode

Historical M3.1 repeated-drain observation

16 / 16 / 1

Three drain batches

Pending returned to zero after every batch

256 / 16

Lifetime events / pending limit

Contract bounds; distinct from observed 33

This is a delivery/liveness scenario with generated evidence. No append latency, sustained ingestion, storage growth or long-duration retention benchmark is recorded.

PROPOSED BENCHMARK · NOT RUN

Proposed: vary append/drain ratio, provenance size and injected relay outage; measure p50/p95 append latency, backlog age and bytes/event without dropping history.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	Private SQLite episode owner over real DeTTa; native episode and inference actions.
FIXTURE / LOCAL TEST POLICY	Generated reports, operator origins/grants and hostile completion responses.
BLOCKED / UNSUPPORTED	No required current gate blocked. Pre-existing M3-format pending work is not repaired by new profiles.
UNRUN	Long-lived production histories, distributed writers, deletion/retention policy and migration.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Measure backlog behavior	Rate-controlled append/drain and outages	No loss; bounded backpressure and recovery age
Add explicit retention policy	Provenance and replay obligations first	Deletion cannot silently invalidate accepted evidence
Improve inspection tools	Authorized read-only views with privacy controls	Explain event order and delivery state without exposing identifiers

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

Processing completion is not the same as numerical agreement.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m3_1.py <env> --output <fresh>
```

Fresh explicit closure profiles; full wrapper includes older regressions. Historical 33-event evidence remains tied to its own tested revision.

`fusion/episodes.py`
`fusion/delivery_contracts.py`
`fusion/evidence_relay.py`
`tests/test_m3_1_delivery.py`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.

BeliefRevision · C2

Admit support carefully.
Own the belief decision.

ELI16 · THE IDEA IN PLAIN LANGUAGE

An evidence ledger: a copied report is still the same source, and a calculation does not erase contrary observations.

Why it matters: Numerical revision is meaningful only when the system tracks which evidence contributed, which assumptions were made and which alternatives remain.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

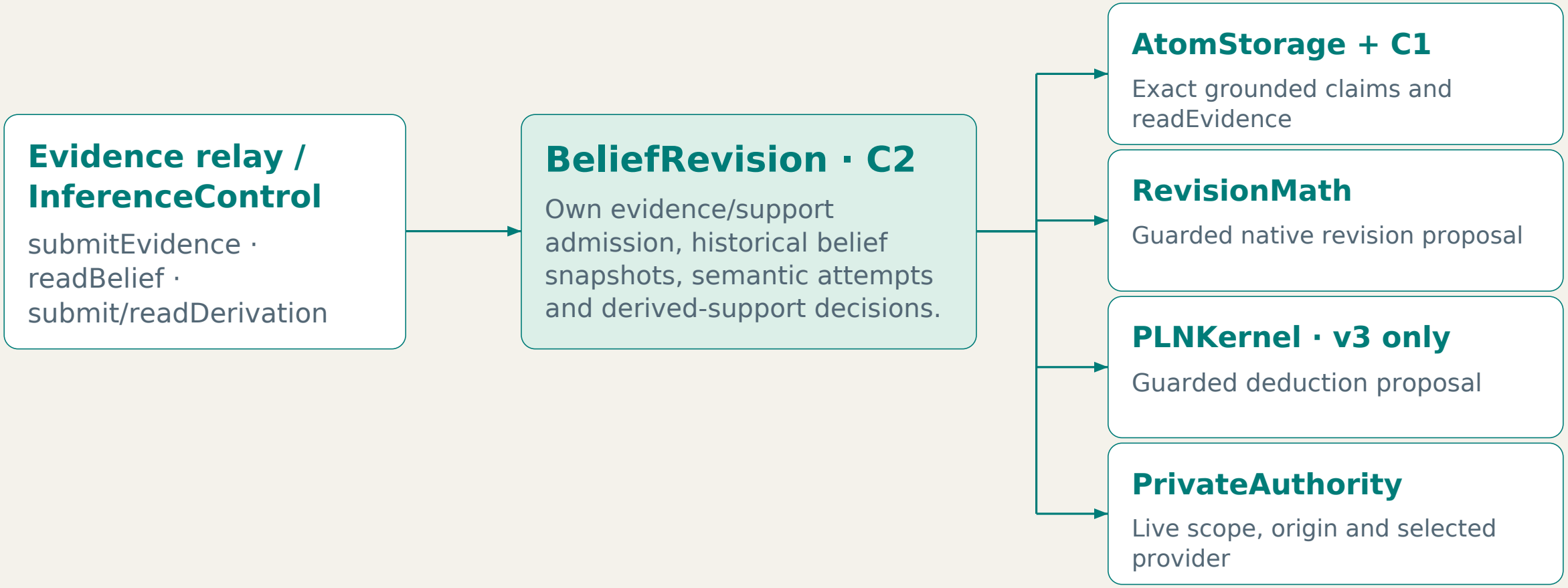
The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

Persistence	Knowledge access	Cognition	Coordination + policy
DA storage	Relations	Episodes · C1	Messaging
AtomStorage · C0	Search projection	**Beliefs · C2**	Execution
ConsensusFinality		PLNKernel	Registry
		Attention · C3	Resolver
		Inference · C4	Authorization

Own evidence/support admission, historical belief snapshots, semantic attempts and derived-support decisions.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Support + observation ledger

Accepted occurrence lineage, alternatives and historical snapshots

Semantic attempts

Exact input bindings and durable admission decisions

OUTSIDE THIS RESPONSIBILITY

Original event history

C1 owns episodes and evidence-delivery intent

Pure math and task scheduling

Providers calculate; Attention ranks; C4 budgets work

M2 pln-revision-private/1; fresh M3 pln-derived-private/1 · local v3

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
submitEvidence	Validate C1 evidence or existing support	Admit/duplicate separately from numerical outcome
readBelief	Read exact historical or current snapshot	Aggregate states its admitted coverage
submitDerivation	Bind exact historical premise vector	v3 tagged support; no new observation
readDerivation	Recover a saved semantic decision	Replay does not force another calculation

Bounded local support history; aggregate covers admitted supports, not every retained observation.

A worked example through the boundary

M2 REAL native PLN + DeTTa · generated observations and explicit independence assumptions

01 · STEP

Retain reports

Store original and contrary observations.

02 · STEP

Check lineage

Copied/ancestor support cannot create independent roots.

03 · STEP

Run guarded revision

Eligible exact pair uses the pinned native formula.

04 · STEP

Commit decision

Keep alternatives and explicit aggregate coverage.

What the example establishes

The owner decides what counts; the calculator only proposes.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Admission ≠ aggregation

A report may be processed without changing a numerical truth.

I2

No double evidence

New key or speaker cannot manufacture an independent origin.

I3

Derived support inherits

Deduction carries original roots and ancestry; no fresh observation count.

I4

Owner commits last

Recheck exact bindings, current authority and expected revision before committing.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Proposal computed; owner commit lost	Recover/retry exact attempt under its budget semantics.	Physical calculation may repeat where allowed
Decision saved; relay reply lost	Return the original decision through declared ports.	Duplicate-safe admission
Stale revision / malformed provider response	Refuse or retain explicit outcome; never silently rebind.	CAS and hostile-response tests

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

0.5 / ⅔

Native revision strength / confidence

Recorded M2 generated example; model-specific

2

Retained alternatives in M2 story

Disagreement remains visible

0 new

Observations from derived support

M3 admission invariant, not an accuracy score

These are numerical goldens and lineage/admission results. They do not measure calibrated confidence, predictive accuracy, source independence or per-admission latency.

PROPOSED BENCHMARK · NOT RUN

Proposed: fix support lineage and premise snapshots; vary duplicate ratio, alternatives and CAS contention. Measure admission p50/p95 and verify unchanged evidence-root accounting.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	Private SQLite owners, actual guarded native revision/deduction and real DeTTa bytes.
FIXTURE / LOCAL TEST POLICY	Generated claims, declared independence assumptions, malformed proposals and injected crashes.
BLOCKED / UNSUPPORTED	No required supported gate blocked. Existing v2/v3 journals are not migrated or silently widened.
UNRUN	Retraction, arbitrary rule libraries, calibration, hidden-copy detection and distributed belief writers.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Measure admission costs	Controlled lineage/duplicate/CAS workloads	Latency reported with duplicate and conflict rates
Specify retraction semantics	Versioned provenance and historical-read contract	No erased history or unaccounted evidence
Evaluate numerical usefulness	External task dataset and calibration protocol	Accuracy/calibration separated from conformance

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

The owner decides what counts; the calculator only proposes.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m3.py <env> --output <fresh>
```

Uses explicit private profiles and pinned native providers. Historical M2 demo is a CLI observation; separate native Hive cases establish the real-caller gate.

`fusion/beliefs.py`
`fusion/derived_beliefs.py`
`fusion/cognition_contracts.py`
`tests/test_m2.py`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.

PLNKernel

Calculate one grounded proposal. Own no beliefs.

ELI16 · THE IDEA IN PLAIN LANGUAGE

A guarded calculator: the same exact inputs enter a pinned formula, and the result goes back to the evidence owner for acceptance.

Why it matters: Separating numerical proposals from admission prevents a runtime or formula replacement from silently acquiring authority over beliefs, budgets or search.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

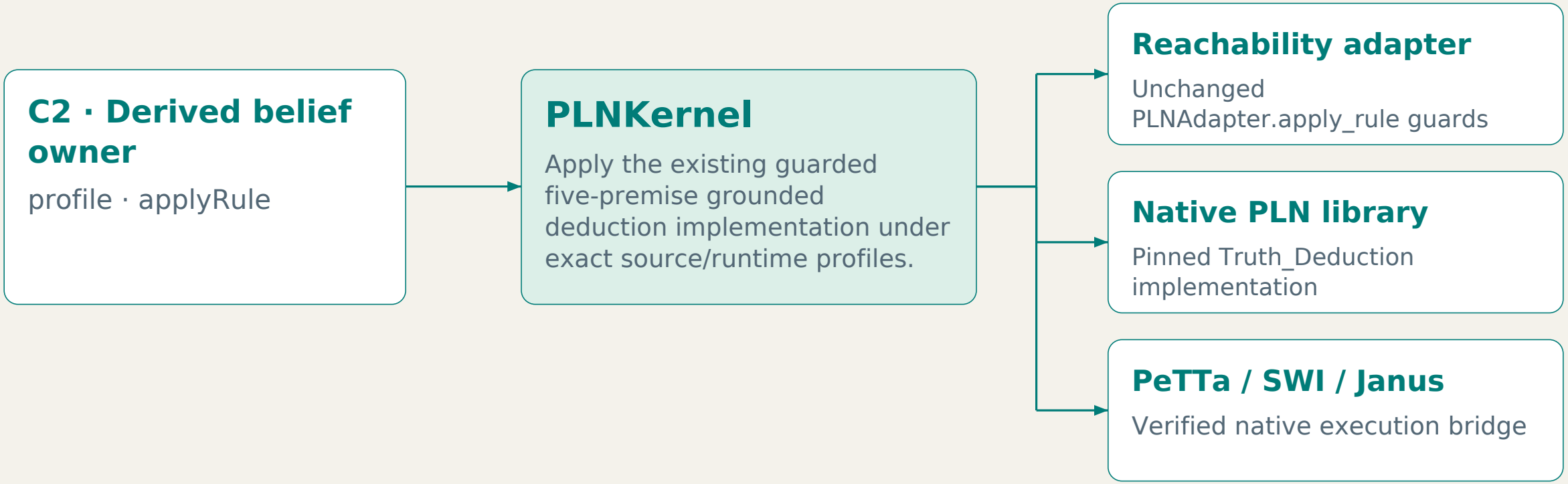
The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

Persistence	Knowledge access	Cognition	Coordination + policy
DA storage	Relations	Episodes · C1	Messaging
AtomStorage · C0	Search projection	Beliefs · C2	Execution
ConsensusFinality		**PLNKernel**	Registry
		Attention · C3	Resolver
		Inference · C4	Authorization

Apply the existing guarded five-premise grounded deduction implementation under exact source/runtime profiles.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

No mutable domain state

A pure proposal interface; selected runtime identity is explicit

Guarded execution boundary

Validate inputs, source/profile compatibility and returned proposal

OUTSIDE THIS RESPONSIBILITY

Admission and provenance authority

C2 validates exact claim/lineage and commits the result

Search and work budgets

No PLN.Query/Derive loop, agenda, task generation or attention mutation

grounded-five-premise-deduction/1 · fusion-petta-swi-10.1.14-deduction/1

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
profile	Identify model, rule, runtime and pinned sources	Compatibility is explicit before use
applyRule	Evaluate P, Q, R, P⇒Q, Q⇒R	One grounded proposal for P⇒R
RevisionMath.revise	Separate narrow revision dependency	Not an extra kernel method or general inference loop

One bounded grounded rule; heuristic truth model; no belief, budget or agenda state.

A worked example through the boundary

REAL native PeTTa / PLN · generated exact premises; no replacement mathematics

01 · STEP

Bind premises

Exact historical supports for five grounded roles.

02 · STEP

Validate

Check domains, bindings and probability consistency.

03 · STEP

Execute native

Run unchanged guarded Truth_Deduction.

04 · STEP

Return proposal

C2 rechecks authority and decides admission.

What the example establishes

Reusing mathematics requires preserving its assumptions and boundary.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Pinned mathematics

No alternative formula is substituted to obtain a desired result.

I2

Exact premise vector

Historical support identities are not replaced by latest beliefs.

I3

Refuse invalid domains

Nonfinite or inconsistent inputs fail guarded checks.

I4

No owner writes

A successful formula result alone changes no belief or attention state.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Native process unavailable	Return explicit unavailable/failed attempt semantics.	No fixture fallback for real gate
Crash around formula return	C2/C4 journals govern recovery and spent attempts.	Kernel is not the recovery owner
Source/runtime mismatch	Reject incompatible selection before admission.	Historical SWI 10.0.1 is unrun

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

5 → 1

Premises → grounded proposal

One rule, not general inference

0.68 / 0.3136

Recorded deduction strength / confidence

Generated golden; tolerance 1e−12

0

Mutable domain owners in kernel

Architectural invariant, not performance

The numerical golden checks implementation agreement, not probabilistic calibration or theorem validity. No isolated native-call latency or large-rule benchmark is recorded.

PROPOSED BENCHMARK · NOT RUN

Proposed: separate process startup from warm native calls; fix exact valid vectors and refusal cases; report p50/p95, guard cost and timeout behavior without changing the formula.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	Pinned Reachability, PLN, PeTTa, SWI 10.1.14 and Janus execute the native formula.
FIXTURE / LOCAL TEST POLICY	Generated premise vectors and independent expected numerical goldens.
BLOCKED / UNSUPPORTED	No required supported gate blocked; historical SWI 10.0.1 compatibility remains unrun.
UNRUN	General rule search, calibration, broad rule coverage and cross-runtime numerical equivalence.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Measure the runtime boundary	Warm/cold fixed-vector experiments	Separate startup, guards and native evaluation
Qualify a new runtime pin	Golden, refusal and source compatibility matrix	No silent numerical or exception changes
Add another rule deliberately	Explicit rule profile and lineage/admission review	Each new rule has independent goldens and refusal tests

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

Reusing mathematics requires preserving its assumptions and boundary.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m3.py <env> --output <fresh>
```

Build pinned providers with build_m3.py first; use a fresh source checkout and explicit runtime selection. The wrapper includes both guard and actual-native suites.

`fusion/adapters/pln_deduction.py`
`integration/pln/test_deduction.py`
`integration/pln/deduction-sources.json`
`docs/module-interfaces/m3-evidence.md`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.

Attention · C3

Rank eligible work.
Credit each cause once.

ELI16 · THE IDEA IN PLAIN LANGUAGE

A priority list with a clock: tasks gain or lose points, but a repeated reward receipt cannot buy a second reward.

Why it matters: Limited work budgets make ordering observable. Keeping attention independent allows policy/provider experiments without changing belief mathematics or task admission.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

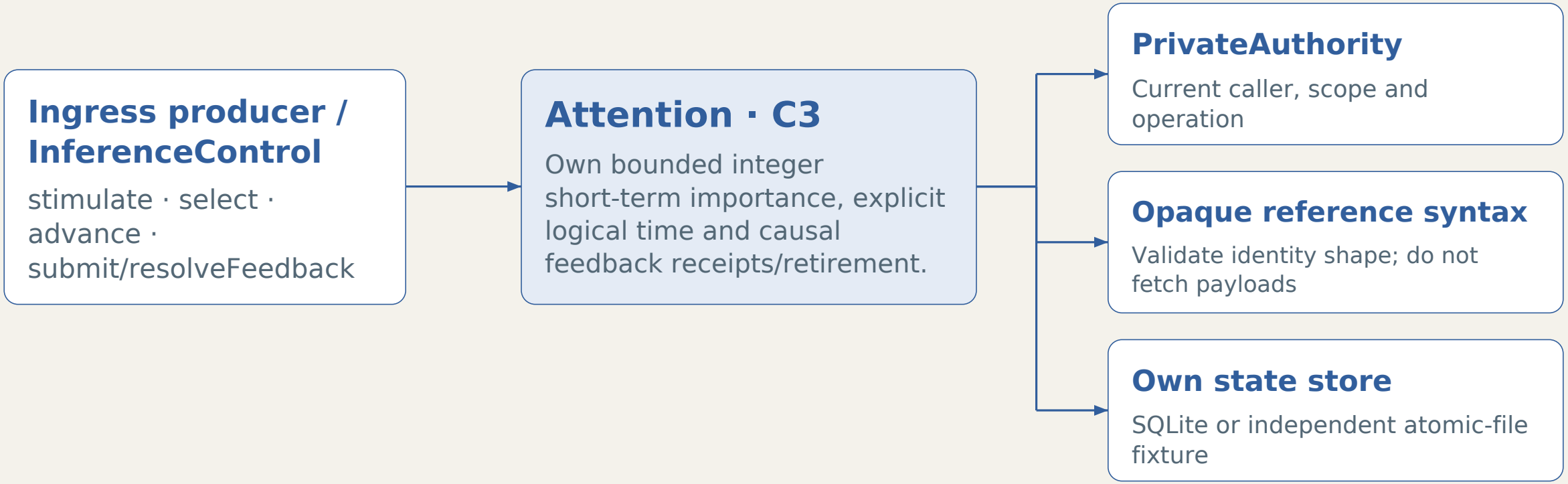
The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

Persistence	Knowledge access	Cognition	Coordination + policy
DA storage	Relations	Episodes · C1	Messaging
AtomStorage · C0	Search projection	Beliefs · C2	Execution
ConsensusFinality		PLNKernel	Registry
		Attention · C3	Resolver
		Inference · C4	Authorization

Own bounded integer short-term importance, explicit logical time and causal feedback receipts/retirement.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Priority and logical clock

STI per target/context, revision and generation watermark

Causal outcome ledger

Command replay, feedback application and no-credit retirement

OUTSIDE THIS RESPONSIBILITY

Truth values

STI is not belief confidence and does not alter formula inputs

Cognition/search

No task creation, ECAN, LTI, wages/funds or hidden inference dependency

sti-additive-linear-decay/1; fresh closure profile sti-additive-linear-decay-closure/1

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
submitStimulus / readAttention	Ingress plus explicit read	Reads do not allocate or decay state
select	Rank positive eligible STI; deterministic tie-break	No discovery of new tasks
advance	Explicit expected-step logical decay	Stale clocks fail; replay does not retime
submitFeedback / resolveFeedback	Apply causal credit or account for prior outcome	Closure cannot newly apply eligible feedback

Default 16 targets/context, 8 contexts; finite cause/command capacity; explicit logical steps.

A worked example through the boundary

HISTORICAL M3 CONTROLLED COMPARISON · SQLite + independent file fixture

01 · STEP

Prefer B

Scores A=10, B=20; one attempt available.

02 · STEP

Select

B runs first; C2 admits one conclusion.

03 · STEP

Credit once

C3 records one causal feedback outcome.

04 · STEP

Replay / swap

Both providers retain outcome; no extra native call.

What the example establishes

Priority selects work; causal receipts prevent repeated reward.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Policy is explicit

Integer saturation, context budget and decay are part of identity.

I2

Causal uniqueness

Replay cannot apply the same logical feedback twice.

I3

Strict logical time

Feedback is never retimed to make stale work admissible.

I4

One authoritative outcome

Application and retirement serialize at the attention owner.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Feedback applied; response lost	Return original application receipt even after expiry/decay.	No second credit
Clock advanced / feedback expired	Retire without credit when permitted.	Do not manufacture application
Eligible/unavailable or finite capacity	Resolution remains pending.	Reconciliation is not a scheduler

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

4

Controlled provider/preference runs

Same policy, tasks and one-attempt budget

1 / 0

Native call / replay calls per run

Whole inference-step observation

100 / 300 / 1

Target cap / context budget / decay

Default STI policy parameters

Historical full-step timings span 0.641150–0.744361 s, one observation per case. They include native inference and cannot rank attention-provider speed or cognitive quality.

PROPOSED BENCHMARK · NOT RUN

Proposed: fix eligible sets and causal schedule; vary target count, replay ratio and feedback contention. Measure select/update p50/p95 while checking exact ranking and once-only outcomes.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	Persistent SQLite reference STI provider; real native inference consumes its selections.
FIXTURE / LOCAL TEST POLICY	Independent atomic-file attention, generated tasks, grants and deterministic fault positions.
BLOCKED / UNSUPPORTED	No required supported gate blocked; old pending stores need their original profile.
UNRUN	ECAN/LTI, learned policy quality, production scaling and populated-state migration.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Benchmark independent providers	Identical logical schedule and durability semantics	Performance without ranking or receipt drift
Evaluate ranking usefulness	Declared task objective and fixed inference budget	Outcome quality separate from conformance
Qualify another policy	New profile with clock/causal semantics	No hidden adoption of LTI or coupled attention state

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

Priority selects work; causal receipts prevent repeated reward.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m3_1.py <env> --output <fresh>
```

Uses fresh closure-profile stores and both providers. Historical comparison retains its own source label; M5 reruns legacy behavior without adding attention work.

`fusion/attention.py`
`fusion/attention_contracts.py`
`fusion/adapters/fixture_attention.py`
`tests/test_m3_1_attention.py`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.

InferenceControl · C4

Spend work deliberately.
Keep the budget spent after a crash.

ELI16 · THE IDEA IN PLAIN LANGUAGE

A limited exam-attempt card: restarting the application does not give back an attempt already reserved.

Why it matters: A bounded native action needs durable limits and honest completion status. Restart must not turn one authorized step into unbounded evaluation.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

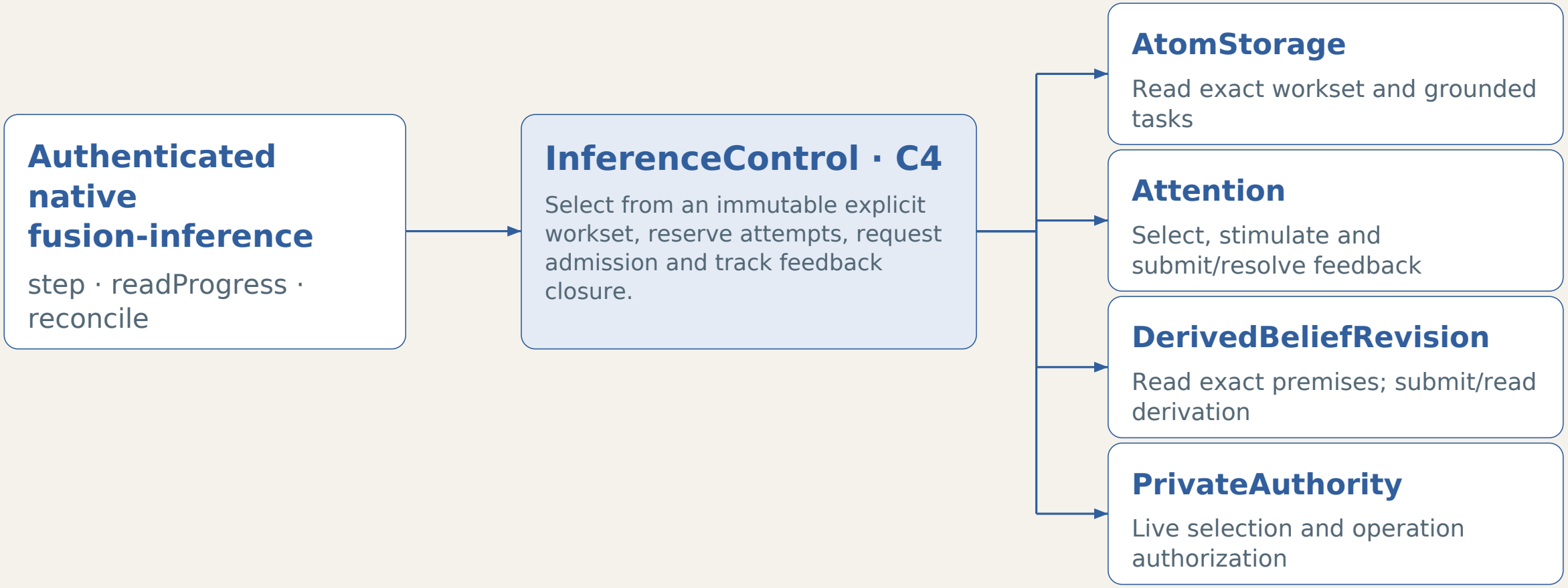
The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

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AtomStorage · C0	Search projection	Beliefs · C2	Execution
ConsensusFinality		PLNKernel	Registry
		Attention · C3	Resolver
		Inference · C4	Authorization

Select from an immutable explicit workset, reserve attempts, request admission and track feedback closure.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Execution journal

Selections, candidate progress, physical-attempt reservations and immutable result

Closure journal

Bounded feedback reconciliation progress, separate from execution counters

OUTSIDE THIS RESPONSIBILITY

Beliefs and STI

Admission belongs to C2; causal credit belongs to C3

General search/scheduling

No agenda expansion, continuous worker or automatic retry scheduler

bounded-grounded-inference/1; fresh M3.1 local 2.1.0 closure extension

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
step	Run one immutable explicit workset	Budget, deadline and selections bind the intent
readProgress	Expose execution and closure independently	Terminal execution may have pending closure
reconcile	Resolve already-recorded feedback intents	No new inference, retiming, refunds or deadline reset

Finite worksets and journals; ≤16 resolutions/reconcile call; absolute deadlines never renewed.

A worked example through the boundary

ACTUAL native Hive inference · generated workset; real C2/PLN/DeTTa

01 · STEP

Freeze selection

Read eligible tasks and exact premise snapshots.

02 · STEP

Reserve

Persist one physical attempt before evaluation.

03 · STEP

Request C2

C2 invokes guarded PLN and decides admission.

04 · STEP

Close separately

Save feedback outcome; later reconcile if needed.

What the example establishes

Bounded work can finish honestly without finding a conclusion.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Durable budgets

Candidate/attempt/generation limits and absolute deadline survive restart.

I2

Physical reservation

A crash may consume a slot without a conclusion; no refund.

I3

Frozen inputs

No silent latest-premise substitution or workset expansion.

I4

Honest progress

Later closure never rewrites the original execution result.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Crash after reservation	Slot remains spent, even if native evaluation never completes.	Conservative bounded-work guarantee
Admission saved; reply lost	Read the recorded C2 decision and continue original intent.	Duplicate-safe support admission
Feedback application reply lost	Resolve at C3; preserve original result and budget.	No new native formula calls

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

2 / 1

Candidates / attempt in comparison

One admission and one feedback credit

0

New native calls during reconciliation

Recorded M3.1 observation

1–16

Tasks in explicit workset

Contract bound, not search depth

The historical four-case comparison records one full step per configuration, not a latency distribution. No sustained scheduling, fairness or general inference benchmark exists.

PROPOSED BENCHMARK · NOT RUN

Proposed: fixed worksets and native runtime; vary budgets, refused candidates and crash points. Report completion/abandonment, physical evaluations and p50/p95 step/closure latency.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	Persistent controller, actual authenticated Hive action and native guarded PLN through C2.
FIXTURE / LOCAL TEST POLICY	Generated tasks/premises, initial priority preference and injected kill positions.
BLOCKED / UNSUPPORTED	No required supported gate blocked. Eligible/unavailable feedback can remain pending by design.
UNRUN	General inference search, scheduler fairness, LTI, distributed recovery and old-state migration.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Measure budget utility	Fixed tasks, declared objective and failure schedule	Useful completions per physical evaluation
Improve closure observability	Read-only authorized receipt tooling	Separate execution, evidence and feedback progress
Specify orchestration separately	Explicit ownership and retry policy before automation	No scheduler hidden inside reconciliation

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

Bounded work can finish honestly without finding a conclusion.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m3_1.py <env> --output <fresh>
```

Actual Hive cases require pinned native providers and private operator configuration. Fresh closure journals preserve explicit compatibility; no live migration.

`fusion/inference.py`
`fusion/deadlines.py`
`fusion/cognition_contracts.py`
`integration/hive/inference_program/program.py`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.

Agent messaging

Accept durably.
Deliver repeatedly until acknowledged.

ELI16 · THE IDEA IN PLAIN LANGUAGE

Registered post: accepting an envelope, showing it to its recipient and receiving a signature are three different events.

Why it matters: A lost reply must not lose accepted work or imply delivery that never happened. Separate mailbox ownership lets execution retry completion safely.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

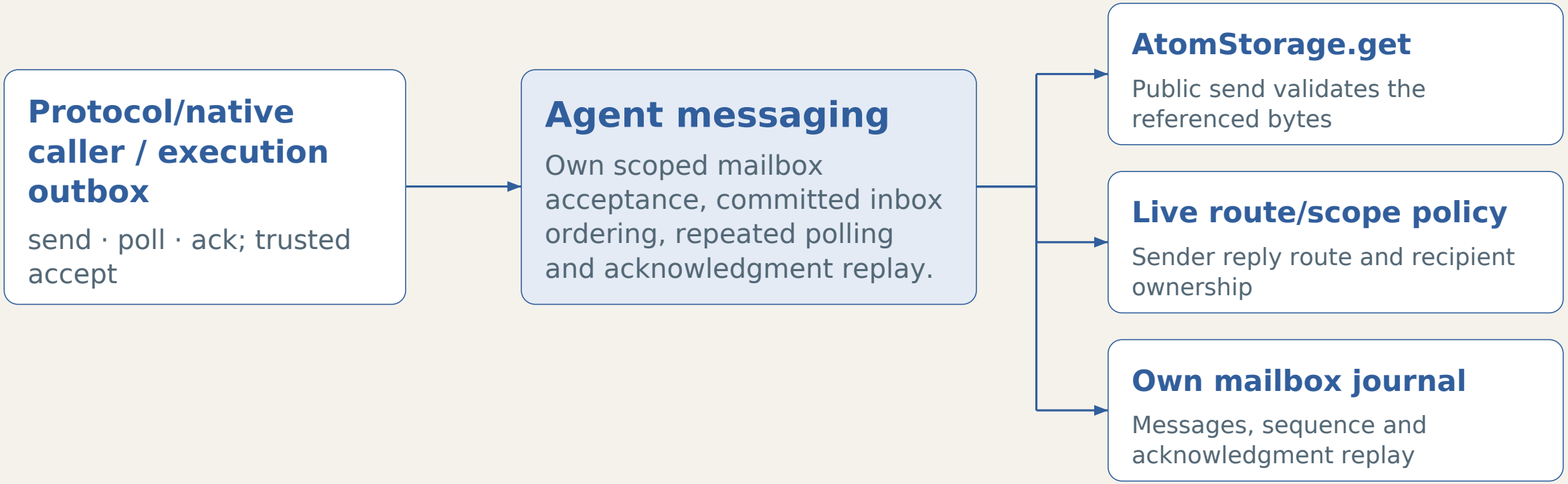
The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

Persistence	Knowledge access	Cognition	Coordination + policy
DA storage	Relations	Episodes · C1	**Messaging**
AtomStorage · C0	Search projection	Beliefs · C2	Execution
ConsensusFinality		PLNKernel	Registry
		Attention · C3	Resolver
		Inference · C4	Authorization

Own scoped mailbox acceptance, committed inbox ordering, repeated polling and acknowledgment replay.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Inbox + ordering

Durable accepted message, scope/route and committed local sequence

Acknowledgment replay

Recipient acknowledgment identity and stable response

OUTSIDE THIS RESPONSIBILITY

Result payloads and jobs

AtomStorage owns bytes; Execution owns terminal job/outbox state

Exactly-once delivery

Poll may repeat; no global ordering or guaranteed remote arrival

durable-mailbox/2 · explicit legacy profile retained · wire 1.0.0

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
send	Validate reference and registered routes; accept	Acceptance reports delivery not-observed
poll	Return oldest unacknowledged message for owner	Repeats until acknowledged
ack	Commit recipient acknowledgment and replay	Current route ownership still checked
Trusted accept	CompletionDelivery.accept: execution-to-mailbox handoff	Return after mailbox commit; not public dispatch

Finite configured unacknowledged-route capacity; replay records retained for journal lifetime.

A worked example through the boundary

REAL owner persistence + separate worker process · protocol counterparty is a fixture

01 · STEP

Finish job

Execution has a durable completion outbox row.

02 · STEP

Accept mailbox

Mailbox commits that completion independently.

03 · STEP

Lose marking

Worker dies before marking its outbox sent.

04 · STEP

Retry handoff

Same message is accepted once; recipient polls and acks.

What the example establishes

A durable receipt must say which delivery fact was observed.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Acceptance before reply

A success receipt follows the mailbox commit.

I2

Repeat until acknowledgment

Poll observation does not consume the delivery.

I3

Scoped recipient control

Only the registered owner can poll/ack its route.

I4

Semantic replay

Same delivery identity/body replays; changed meaning conflicts.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Mailbox accepted; outbox not marked	Execution retries trusted accept with the same identity.	No duplicate logical message
Acknowledgment reply lost	Retry recovers durable acknowledgment after reauthorization.	No resurrection of consumed delivery
Full / revoked reply route	Outbox remains pending until a valid retry can succeed.	No dropped completion

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

2

Messages in the preserved inspect workflow

One reference share + one completion

0

Pending outbox after recovery demos

Observed logical handoff completion

FIFO

Committed inbox sequence per route

Ordering contract, not global message order

These are delivery/replay observations from shared workflows. No isolated mailbox throughput, delivery p95, multi-consumer scale or network-loss benchmark is recorded.

PROPOSED BENCHMARK · NOT RUN

Proposed: vary backlog, ack delay, replay ratio and route capacity; measure send/poll/ack p50/p95, backlog age and duplicate physical deliveries under fixed semantics.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	SQLite mailbox owner and actual native caller binding; real process restarts.
FIXTURE / LOCAL TEST POLICY	Protocol peer/counterparty, generated messages, local route policy and injected faults.
BLOCKED / UNSUPPORTED	No required current gate blocked; a full or revoked route intentionally delays completion.
UNRUN	Two production agents, remote brokers, distributed ordering, power loss and general messaging federation.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Measure backlog and replay costs	Rate-controlled sends/acks with fixed capacity	No loss while reporting tail latency and queue age
Qualify a remote transport	Preserve authority, acceptance and acknowledgment meanings	Failure injection across actual network partitions
Specify retention/compaction	Replay lifetime and acknowledgment contract first	No duplicate acceptance after tombstone expiry

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

A durable receipt must say which delivery fact was observed.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m11.py
```

Runs the provider/binding conformance matrix and real DeTTa gate when available. Fixture-only is a distinct mode; it is not native Hive proof.

`fusion/messaging.py`
`fusion/execution.py`
`tests/test_contract.py`
`tests/test_m11.py`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.

Delegated execution

Own the job and its outbox. Let other owners commit their effects.

ELI16 · THE IDEA IN PLAIN LANGUAGE

A service desk with a job ticket: “accepted,” “finished” and “customer notified” have separate receipts.

Why it matters: Delegated work crosses storage and messaging boundaries. Explicit jobs and an outbox allow recovery without a transaction spanning all three owners.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

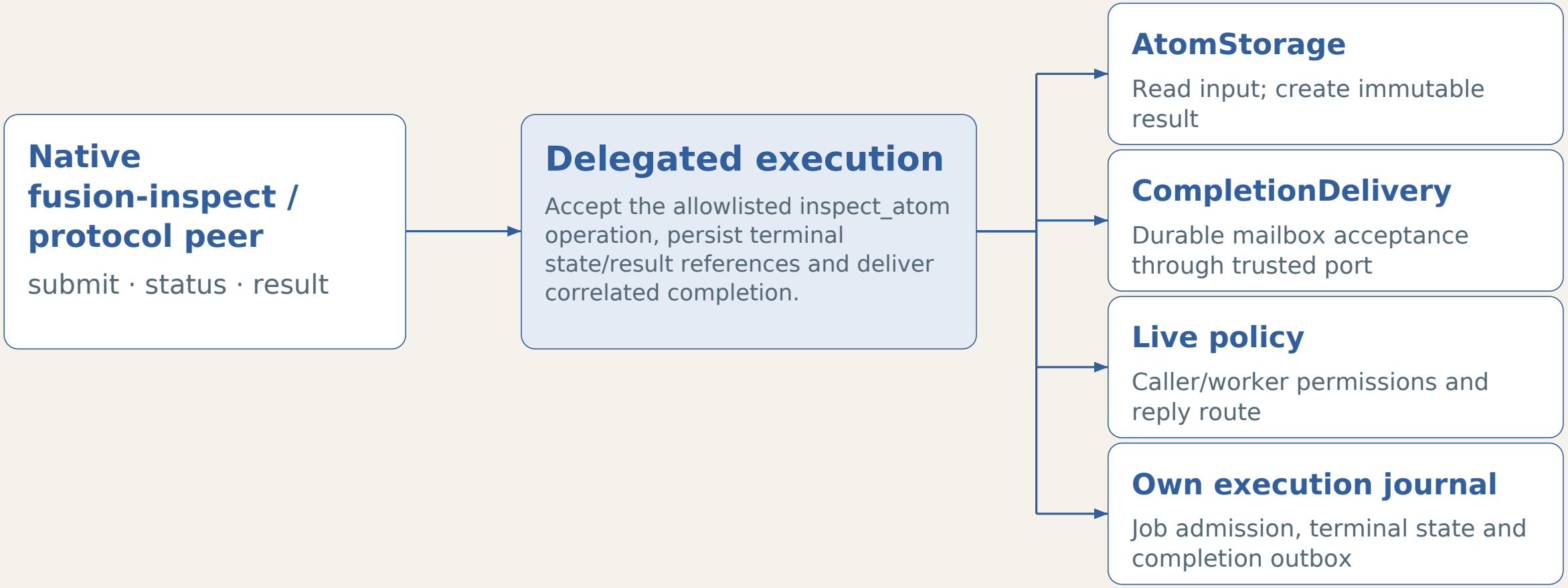
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ConsensusFinality		PLNKernel	Registry
		Attention · C3	Resolver
		Inference · C4	Authorization

Accept the allowlisted inspect_atom operation, persist terminal state/result references and deliver correlated completion.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Job + terminal decision

Queued/succeeded/failed and result/error correlation

Completion outbox

Delivery intent and recorded mailbox-acceptance progress

OUTSIDE THIS RESPONSIBILITY

Storage or mailbox mutation authority

Calls declared ports; no foreign tables or cross-owner SQL transaction

Reasoning/process migration

Only inspect_atom; no arbitrary code execution or cognition

inspect-atom/2 · operation algorithm 0.1.0 · public wire 1.0.0

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
submit	Validate caller/worker read permissions and input	Durable queued acceptance, not completion
status	Read caller-owned job and delivery observations	Execution and delivery remain distinct
result	Return accepted result reference after success	Reauthorize read; failed/pending outcomes explicit
Trusted worker / flush	Compute bounded result, persist and deliver	Not exposed as arbitrary client commands

One bounded worker writer; finite configured queued-job + unsent-outbox capacity.

A worked example through the boundary

ACTUAL native Hive inspect + real DeTTa · deterministic worker; fixture counterparty

01 · STEP

Accept input

Read the generated atom and accept inspect_atom.

02 · STEP

Persist result

Store digest, byte length and algorithm version.

03 · STEP

Commit terminal

Save result reference + outbox in one owner commit.

04 · STEP

Deliver / recover

Mailbox accepts; caller verifies and acknowledges.

What the example establishes

Execution completion and completion delivery have different owners.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Allowlisted work

Only the named operation is accepted; no caller-supplied code.

I2

Result before terminal

A terminal success references already persisted result bytes.

I3

Atomic outbox intent

Terminal state and completion intent commit together.

I4

Live authority/deadline

Restart rechecks caller/worker permissions; no new work after start deadline.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Result persisted; terminal commit lost	Retry immutable result creation, then commit terminal/outbox.	Stable result identity
Mailbox accepted; sent flag lost	Retry handoff through CompletionDelivery.	Mailbox owns deduplication
Unavailable input / revoked worker	Keep recoverable work or record explicit terminal failure as specified.	No unauthorized success

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

1 / 1

Job / outbox in preserved workflow

Stable across recovery matrix

143 B

Generated M4 result payload

Exact accepted artifact, not a throughput figure

13

Original native inspect cases

Preserved M1.2 suite; not 13 unique new M5 cases

The fixed inspect operation demonstrates orchestration and recovery, not compute efficiency. No standalone job-admission latency, worker throughput or fairness benchmark is recorded.

PROPOSED BENCHMARK · NOT RUN

Proposed: vary input size, queue depth and downstream failure; measure admission/result/notification separately, including duplicate physical attempts and pending outbox age.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	Local persistent job/outbox, separate worker process, actual native Hive and real DeTTa.
FIXTURE / LOCAL TEST POLICY	Generated inspect workload, protocol counterparty, local grants and deterministic fault placement.
BLOCKED / UNSUPPORTED	No required current gate blocked; unavailable input or full/revoked route may leave progress pending.
UNRUN	Arbitrary workloads, remote worker scheduling, distributed failover and exactly-once physical execution.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Benchmark each lifecycle stage	Independent timestamps for acceptance/result/delivery	Do not report notification delay as execution time
Add an operation explicitly	Determinism, resource bounds and authorization review	New conformance and crash boundary tests
Explore remote workers	Placement and failure contract before transport	Keep independent owner commits and stable replay

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

Execution completion and completion delivery have different owners.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m11.py
```

For actual Hive, use the pinned M1.2 build/run recipe or the current full legacy verifier; the M1.1 command alone is not a native-caller gate.

`fusion/execution.py`
`fusion/service.py`
`tests/test_m11.py`
`integration/hive/test_integration.py`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.

Module registry

Describe what is configured.
Do not turn a claim into authority.

ELI16 · THE IDEA IN PLAIN LANGUAGE

A catalogue of equipment labels: it says which instrument is installed and what it advertises, not whether you may use it or whether every claim is proven.

Why it matters: A consumer must select semantics rather than a provider name. Explicit descriptors expose compatibility, placement and evidence gaps before effects occur.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

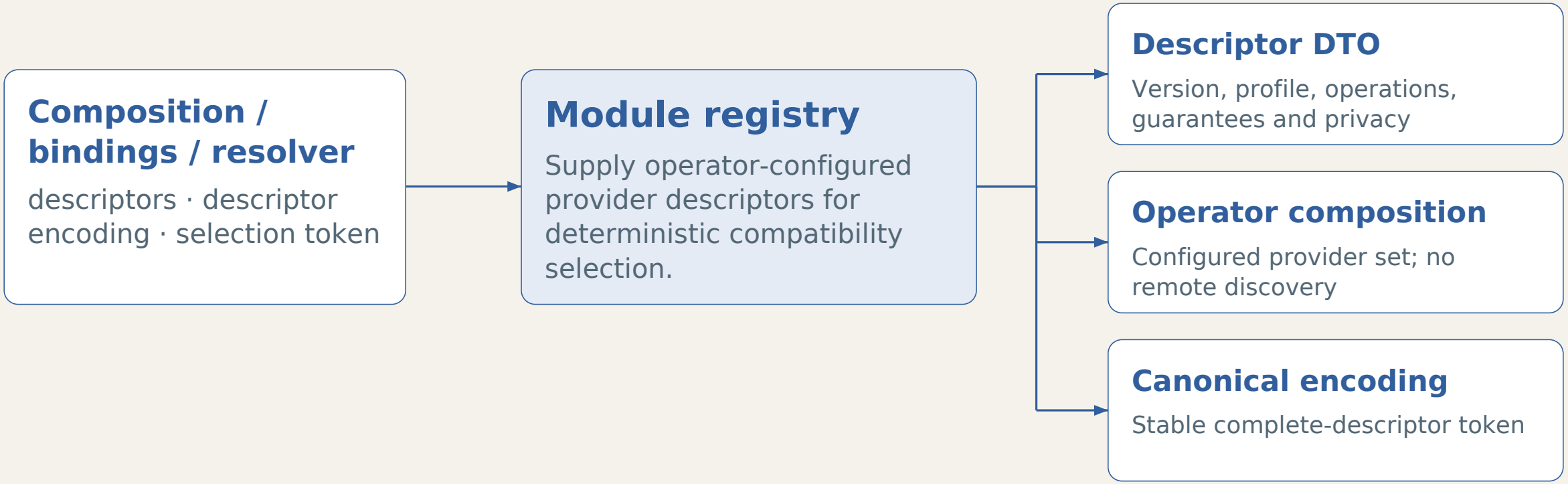
The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

Persistence	Knowledge access	Cognition	Coordination + policy
DA storage	Relations	Episodes · C1	Messaging
AtomStorage · C0	Search projection	Beliefs · C2	Execution
ConsensusFinality		PLNKernel	**Registry**
		Attention · C3	Resolver
		Inference · C4	Authorization

Supply operator-configured provider descriptors for deterministic compatibility selection.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Configured descriptor list

Provider identity, implementation version and advertised contract metadata

No runtime domain journal

No job, belief, balance, registration transaction or mutable remote catalogue

OUTSIDE THIS RESPONSIBILITY

Permission and proof

A descriptor grants no capability and proves no advertised guarantee

Dynamic discovery

No gossip, chain registration, health scheduler or public mutation API

Static catalogue responsibility · static/1; descriptor and implementation versions are separate

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
descriptors	Return configured M1 provider descriptors	Independent of transport; selected AtomStorage provider explicit
descriptor_wire / descriptor_read	Encode/decode typed metadata	Preserve supported fields and sets
selection_token	Bind the complete canonical descriptor	Configuration consistency, not remote authentication
Composition catalogues	Supply private cognition/lookup selections	Explicit profile requirements, not automatic inference

Trusted operator configuration; no independent verification of provider claims.

A worked example through the boundary

LOCAL IMPLEMENTATION · configuration and compatibility fixtures; no registry network

01 · STEP

Configure

Operator installs DeTTa-backed AtomStorage.

02 · STEP

Describe

Publish version, operations, profile and local durability.

03 · STEP

Select

Resolver matches a fixed consumer requirement.

04 · STEP

Bind

Invocation checks the complete selected token again.

What the example establishes

A useful descriptor makes both capabilities and uncertainty visible.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Description ≠ permission

Authorization remains a separate current decision.

I2

Description ≠ evidence

Advertised guarantees or measured_slo fields are not attestations.

I3

Complete identity

Changing any selected descriptor field changes the binding token.

I4

Transport neutrality

Selecting stdio vs in-process does not redefine semantics.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Operator changes configured descriptor	An existing binding fails closed on token mismatch.	No silent provider switch
Descriptor advertises unsupported feature	Resolver refuses the incompatible requirement.	Explicit compatibility error
Process restart	Re-read operator configuration; old caller intent still pins selection.	No registry state recovery transaction

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

3

M1 provider descriptors

AtomStorage, Messaging and Execution catalogue

2 × 2

Storage provider/binding matrix

Same consumer selection semantics exercised

0

Proof verifiers in registry

Required verification is not supplied by advertising

No catalogue lookup latency, update propagation or large-provider-count benchmark exists. Descriptor counts are implementation scope; conformance tests check consistency and refusal.

PROPOSED BENCHMARK · NOT RUN

Proposed: fixed valid/invalid descriptor corpora at increasing catalogue size; measure encode/decode/token costs and memory, separately from resolver selection.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	Executed local descriptor construction/encoding and composition code.
FIXTURE / LOCAL TEST POLICY	Operator configuration, advertised metadata and synthetic compatibility descriptors.
BLOCKED / UNSUPPORTED	Verified remote guarantees cannot be satisfied: no proof verifier is implemented.
UNRUN	Remote discovery, dynamic registration, health monitoring, signed attestations and registry scaling.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Make metadata evidence-backed	Verifier identity and evidence expiry contract	Rejected stale or unverified claims stay visible
Improve operator inspection	Read-only catalogue and selection explanation	Show why a provider is installed versus selected
Explore dynamic catalogues	Ownership, consistency and revocation semantics first	No name-only or stale-token fallback

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

A useful descriptor makes both capabilities and uncertainty visible.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m11.py --fixture-only
```

Fixture-only validates registry/resolver/binding behavior without claiming real DA, Hive or chain. Private compositions have additional explicit descriptor checks in their own suites.

`fusion/registry.py`
`fusion/bindings.py`
`fusion/cognition.py`
`fusion/lookup_composition.py`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.

Compatibility resolver

Select matching semantics. Fail before effects.

ELI16 · THE IDEA IN PLAIN LANGUAGE

An adapter checklist: matching the plug shape is not enough; voltage, capacity and required guarantees must match too.

Why it matters: Provider substitution is safe only when consumers state requirements independently and mismatches cannot quietly weaken behavior.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

Persistence	Knowledge access	Cognition	Coordination + policy
DA storage	Relations	Episodes · C1	Messaging
AtomStorage · C0	Search projection	Beliefs · C2	Execution
ConsensusFinality		PLNKernel	Registry
		Attention · C3	**Resolver**
		Inference · C4	Authorization

Pure deterministic filtering of descriptors against explicit version, profile, operation, guarantee and resource requirements.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

No mutable domain state

A pure selection function; callers retain chosen descriptor identity

Compatibility decision

Filter/error ordering and deterministic tie-break

OUTSIDE THIS RESPONSIBILITY

Execution or liveness

Selecting a descriptor does not prove the provider is reachable or correct

Evidence verification

Required verification currently fails; no verifier is implemented

same-major/1 responsibility; explicit exact profiles; M1 wire remains 1.0.0

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
Version filter	Same major and explicit [minimum, maximum) range	No candidate → UNSUPPORTED_VERSION
Profile/features filter	Exact profile and required feature subset	Mismatch → UNSUPPORTED_PROFILE
Operation filter	Required operation subset	Mismatch → UNSUPPORTED_OPERATION
Guarantee/resource filter	Guarantees, payload, privacy and latency requirement	Unsatisfied → NOT_AVAILABLE
Deterministic selection	Highest matching version/provider ordering	No hidden semantic relaxation

Trusted-local metadata checks; no proof of advertised reliability or measured performance.

A worked example through the boundary

EXECUTED LOCAL RESOLUTION · positive/negative descriptors are fixtures

01 · STEP

State need

Consumer requires scoped, idempotent process restart.

02 · STEP

Reject mismatch

Wrong profile, missing operation or weak guarantee fails.

03 · STEP

Choose match

Resolve explicit provider and profile.

04 · STEP

Check again

Binding verifies actual token before an effect.

What the example establishes

Compatibility is a conjunction of requirements, not a matching name.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Independent requirement

Do not derive the required guarantees from whichever provider appears.

I2

Exact semantics

Same major alone is insufficient; profile/features/operations must match.

I3

Fail-closed proof need

An unsupported required verifier cannot become optional.

I4

No silent drift

Changed selected provider/token is refused by the binding.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Provider configuration changes after selection	Invocation rejects descriptor drift.	Caller must select explicitly again
No eligible candidate	Return a typed compatibility error; create no domain effect.	No weaker fallback
Restart with saved intent	Use the pinned selection in that intent.	Fresh replacement needs explicit composition

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

5

Ordered compatibility stages

Version → profile → operations → resources → selection

1.2.3

Additive same-major fixture

Resolver test; actual M1 wire stays 1.0.0

0

Runtime effects from resolution

Pure-function boundary

These are deterministic conformance properties, not measured selection performance. Advertised latency metadata is not a benchmark or verified SLO from this resolver.

PROPOSED BENCHMARK · NOT RUN

Proposed: increasing descriptor count and rejection-stage mixes; measure p50/p95 selection time and allocations, while preserving exact error and tie-break outcomes.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	Executed pure Python resolution and descriptor-bound in-process/stdio calls.
FIXTURE / LOCAL TEST POLICY	Synthetic versions, profiles, guarantees, payload limits and SLO metadata.
BLOCKED / UNSUPPORTED	A requirement for verification cannot pass until a supported verifier exists.
UNRUN	Large dynamic catalogues, remote attestation, availability optimization and adaptive scheduling.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Explain selection decisions	Structured reason trace without relaxing filters	Operators can reproduce each rejection
Connect verified evidence	Explicit verifier and freshness contract	Only independently checked claims satisfy proof requirements
Measure catalogue scale	Controlled sizes and worst-case rejection mix	Deterministic semantics survive optimization

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

Compatibility is a conjunction of requirements, not a matching name.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m11.py --fixture-only
```

Positive and negative descriptor fixtures are the appropriate evidence here; they do not validate a provider’s real-world SLO or security.

`fusion/registry.py`
`fusion/bindings.py`
`tests/test_contract.py`
`tests/test_m11.py`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.

ConsensusFinality

Commit the approved result. Verify its actual on-chain effect.

ELI16 · THE IDEA IN PLAIN LANGUAGE

A signed sealed envelope: retry the same envelope after a lost reply, then read the ledger to check what was actually stored.

Why it matters: Submission, inclusion and finalized effect are different observations. Exact approval and durable signed intent prevent retries from changing the authorized artifact.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

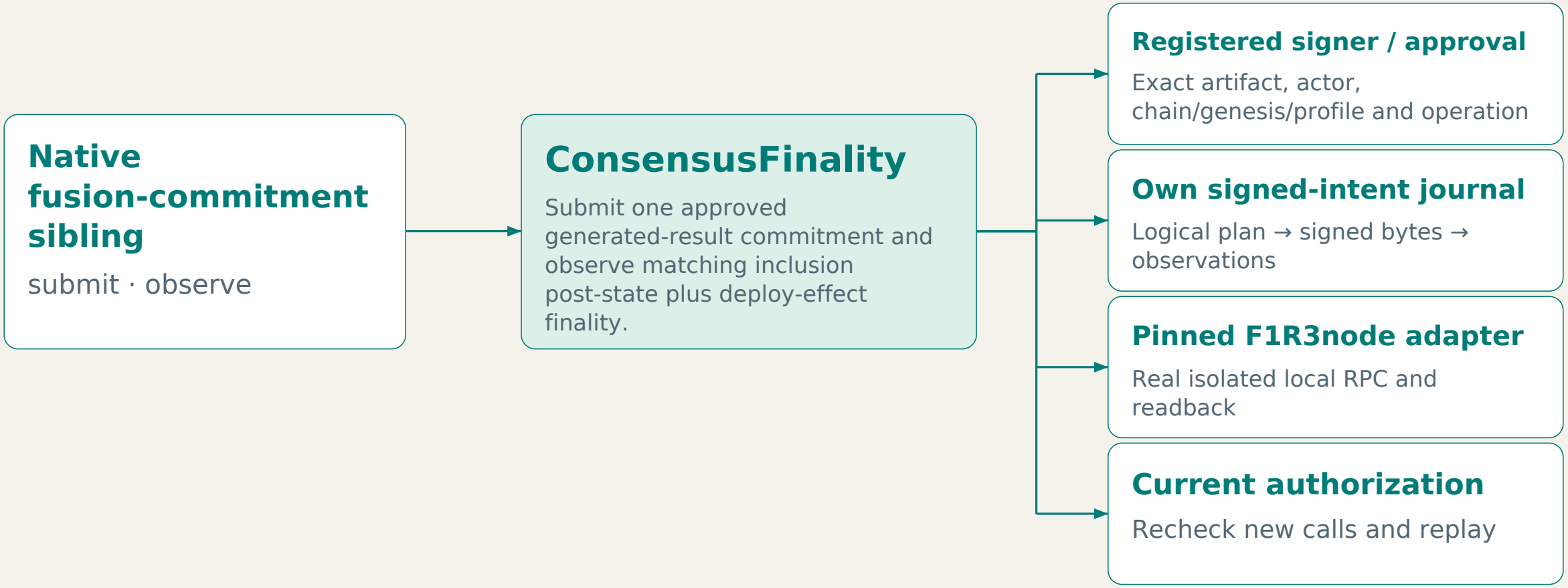
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ConsensusFinality		PLNKernel	Registry
		Attention · C3	Resolver
		Inference · C4	Authorization

Submit one approved generated-result commitment and observe matching inclusion post-state plus deploy-effect finality.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Logical and signed intents

Stable timestamp, deployment identity and full signed request

Verified observations

Exact approved tuple, inclusion block/post-state and matching deploy effect

OUTSIDE THIS RESPONSIBILITY

Consensus implementation

No rewritten consensus, public network or new M5 chain operation

Cognitive correctness

Finality does not prove reasoning, belief truth or off-chain retention

f1r3-generated-post-state/2 · clean F1R3node 9ec1096 · fresh journal

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
submit	Validate approval; persist plan before signing, bytes before send	Replay reuses the exact signed deployment
observe	Check inclusion, post-state value and deploy-effect finality	Finalized block alone is insufficient
Independent readback	Read same effect with read-only access	Separate process; no caller journal or signing key

One local validator; bounded call-driven observation, no public-network attachment.

A worked example through the boundary

M4 REAL isolated F1R3node · generated workflow, test genesis/key and fixture counterparty

01 · STEP

Accept result

Real Hive inspect finishes; retrieve approved bytes.

02 · STEP

Freeze approval

Bind exact result, destination, signer and operation.

03 · STEP

Persist / send

Save intent and signed bytes, then submit locally.

04 · STEP

Verify effect

Read inclusion post-state and matching finalized deploy.

What the example establishes

Verify the approved effect, not merely a successful submission.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Exact-artifact approval

Mutable aliases cannot substitute another result or destination.

I2

Same signed replay

Lost responses do not generate a new timestamp, signature or deployment.

I3

Effect-specific finality

Accepted tuple and successful target deploy effect must match.

I4

Private placement

Episodes, beliefs, provenance and identifying hashes stay off-chain.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Caller dies after result acceptance	Recover accepted artifact, reauthorize and continue.	No substitute artifact
Adapter dies before/after sending	Replay persisted signed bytes, even without signer access.	Four total M4 caller/adapter kill boundaries
Node restarts / observation reply lost	Observe same genesis/state; independent reader confirms effect.	No claim of decentralized failover

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

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First finalized-effect observation

One M4 local run; includes proposal/observation overhead

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Persisted signed request

Same bytes survive replay

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Recorded execution cost

Local fixed workflow; not an economic estimate

Single local validator and one generated artifact, with concurrent regressions. No throughput distribution, public-network latency, economic cost or decentralized-safety benchmark is established.

PROPOSED BENCHMARK · NOT RUN

Proposed: repeat the same isolated workload with fixed genesis/config; report inclusion and effect-finality p50/p95 separately, signed size and execution cost, including outage cases.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	Pinned F1R3node, actual Hive/DeTTa artifact, post-state query and independent readback.
FIXTURE / LOCAL TEST POLICY	Generated input, test signer/genesis, local approval policy and message counterparty.
BLOCKED / UNSUPPORTED	Exact dirty historical M1 source reproduction remains blocked; clean M4 rebaseline passes separately.
UNRUN	Public networks, multi-validator safety, production keys/economics, power loss and trustless attestation.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Measure repeated local behavior	Fixed isolated config and repeated artifact trials	Inclusion and effect-finality distributions separated
Strengthen observation trust	Explicit independent-proof and threat-model contract	Do not confuse node assertions with trustless attestation
Evaluate broader deployment	Separate authorization, isolation and multi-validator study	No silent promotion of local proof into network safety

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

Verify the approved effect, not merely a successful submission.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m4.py <build> --output <fresh>
```

Use the historical tested M4 checkout 55ed540 or its intended compatible scope. M5 verifies unchanged real-chain behavior via verify_m5.py; source, namespace and recipe requirements remain explicit.

`fusion/adapters/f1r3_m4.py`
`integration/hive/commitment/program.py`
`evidence/m4/final/same-artifact.json`
`evidence/m4/final/readback.json`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.

Authorization

Trust the caller context. Recheck authority on replay.

ELI16 · THE IDEA IN PLAIN LANGUAGE

A venue badge checked at each controlled door: a past entry receipt does not restore permission after the badge is revoked.

Why it matters: Replay, cached receipts and provider selection must not bypass current scope, method, route or approval policy. Identity must come from trusted authentication.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

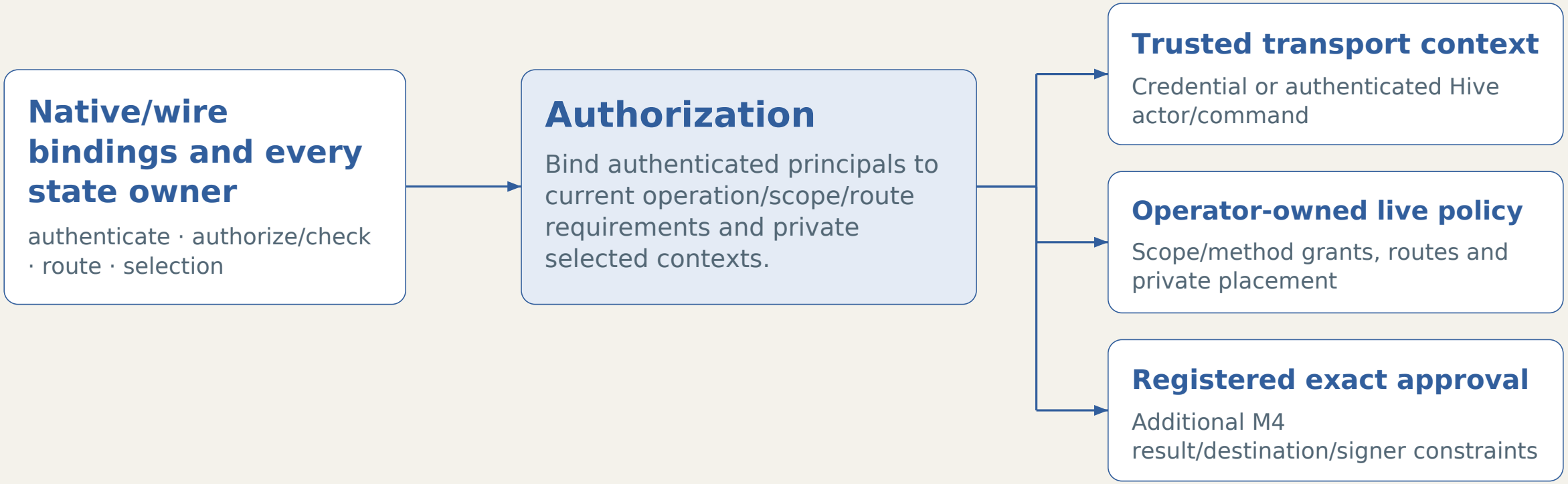
The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

Persistence	Knowledge access	Cognition	Coordination + policy
DA storage	Relations	Episodes · C1	Messaging
AtomStorage · C0	Search projection	Beliefs · C2	Execution
ConsensusFinality		PLNKernel	Registry
		Attention · C3	Resolver
		Inference · C4	**Authorization**

Bind authenticated principals to current operation/scope/route requirements and private selected contexts.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Current policy configuration

Credentials, grants, routes, origin mappings and selected contexts

No business-effect journal

Each domain owner checks authority around its own effects and replay

OUTSIDE THIS RESPONSIBILITY

Caller-selected identity

No sender field, scope name or parameter bundle can grant itself authority

Production identity system

No claim of remote attestation, secrecy from administrators or full IAM

local-test-policy/1; private authority/selection ports; exact M4 approval constraints

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
authenticate	Validate credential; derive registered principal	Trusted transport, not user body identity
check / authorize	Require current scope and operation grant	Recheck cached replay and sensitive owner commits
route	Validate scope and route owner	Only owner polls/acks; worker route checked
Private selection / approval	Bind profile, placement and exact eligible result	Descriptor consistency is separate from permission

Trusted local host/operator; directory placement is not encryption or administrator secrecy.

A worked example through the boundary

REAL Hive authentication · explicitly local operator policy and generated grants

01 · STEP

Authenticate

Hive supplies actor and command from native context.

02 · STEP

Bind privately

Operator mapping selects scope, credential and provider.

03 · STEP

Revoke

Policy changes while an intent is pending.

04 · STEP

Replay refuses

Saved success does not bypass current authority.

What the example establishes

A valid old receipt can still require a current denial.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Trusted identity source

The request body never overrides authenticated principal.

I2

Current authority

Replay is not an authorization cache.

I3

Least owner access

Scope/method/route checks remain at declared boundaries.

I4

Private-by-placement

Public receipts/log parameters avoid private values and identifying hashes.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Worker permission revoked before restart	Recheck at work/recovery; no unauthorized result success.	Legacy worker-revocation tests
Route revoked after job completion	Retain outbox until authorized delivery can proceed.	No dropped completion
Private/chain approval changed before replay	Reject altered selection or ineligible intent.	Current checks and exact-result refusal tests

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

4

Worker-revocation matrix cases

Two storage providers × two bindings, historical focused recheck

LIVE

Policy read on current checks

Implementation behavior; no cached grant

NO SLO

Production authorization benchmark

Security properties require adversarial validation, not only speed

Current evidence is refusal, revocation, canary/privacy and trusted-identity testing. No production attack audit, authorization latency distribution or remote security certification exists.

PROPOSED BENCHMARK · NOT RUN

Proposed: measure authenticate/check/route p50/p95 across policy size and revocation churn; pair every timing run with unauthorized/replay/identity-spoof refusal checks.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	Authenticated native Hive dispatch and executed current-policy checks in local owners.
FIXTURE / LOCAL TEST POLICY	Local test credentials/grants/routes, operator mappings, canaries and malformed requests.
BLOCKED / UNSUPPORTED	No required supported gate blocked; absent grants or revoked access are deliberate denials.
UNRUN	Production IAM, adversarial host compromise, remote attestation, distributed revocation and security audit.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Harden identity integration	Explicit production threat model and trust roots	No body-field impersonation or stale-grant replay
Measure revocation behavior	Controlled churn with concurrent reads/retries	Denial timing and false-acceptance checks together
Add audit views carefully	Authorized redacted event policy	Explain denials without leaking private data or credentials

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

A valid old receipt can still require a current denial.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m5.py --environment <env> --chain-build <chain> --output <fresh>
```

The complete supported regression covers current authority across legacy and new lookup paths. Native authentication and local policy remain separately labelled.

`fusion/policy.py`
`fusion/private.py`
`fusion/service.py`
`tests/test_m11.py`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.